1. Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.



- **Pin 1:** White/Orange
- **Pin 2:** Orange
- **Pin 3:** White/Green
- **Pin 4:** Blue
- **Pin 5:** White/Blue
- **Pin 6:** Green
- **Pin 7:** White/Brown
- **Pin 8:** Brown

2. List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Cable Type	Real-World Device / App	Why It Is Used?
Cat5e	Zomato POS Terminal	A point-of-sale (POS) terminal only needs a stable connection to send small amounts of text data (like orders and bills). Cat5e is cheap, easy to install, and offers more than enough speed for basic office or restaurant machines.
Cat6	Home Wi-Fi Router	Modern home internet plans can reach speeds up to 1 Gbps. Cat6 cables handle these higher speeds smoothly and reduce interference, making sure your smart TVs, gaming consoles, and phones get a fast, lag-free connection from the router.
Fiber Optic	IPL Streaming Server	Streaming live sports to millions of viewers at the same time requires massive bandwidth and zero delays. Fiber optic cables use light to send huge amounts of data across massive distances instantly, making them perfect for major data centers and streaming servers.

3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

(a) measure about 1 to 2 inches from the end of the ethernet cable. Use the stripping blade o the crimping tool to gently score the outer plastic jacket without cutting into the inner wires. Peel away and remove the jacket to expose the four twisted pairs.

(b) Separate and untwist the 4 pairs of wires. Straighten them out and arrange them into the correct color coding order.

(c) Hold the ordered wires tightly between your thumb and forefinger. Use scissor or the cutting blade on the crimper to cut the wire ends straight and even, leaving about half inch of exposed wire sticking out.

(d) Carefully slide the aligned wires into the RJ-45 plug with the gold pins facing upward. Ensure that each wire enters its own separate channel and pushes all the way to the front tip of the connector. And make sure the outer cable jacket extends slightly inside the base of the connectors so the plastic tab can clamp down on it.

(e) Place the RJ-45 plug firmly into the 8 pin crimping slot on the crimp tool. Squeeze the handle firmly all the way down until the ratchet releases. This pushes the metal pins down to bite into the copper wires and locks the strain-relief tab onto the cable jacket.

(f) Visually inspect the connector to ensure all wires reach the front and the copper pins are fully depressed. Finally, plug both ends of the cable into the network cable tester to verify that all 8 wires have a solid connection.

2- common mistakes to avoid.

(a) **Stripping too much :** leaves individual coloured wires exposed outside the connector, making them vulnerable to damage, interference or snapping.
(b) **Stripping too little:** causes the plastic stain relief of the tab of the connector to clamp down onto the individual wires instead of the thick outer jacket making the connection weak and prone to pulling apart easily.
(c) **Incorrect color order:** Failing to follow the correct colour order or mixing up either end will result in a crossed or failed connection, preventing the cable from proper working

4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

(a) **For the Live IPL Screens (From the Internet Source to the Event Area)**
Cable: Fiber Optic Cable
Connector: LC (Lucent Connector) or SC (Standard Connector)
Why: Live streaming an IPL match in high quality requires massive amounts of data. Fiber optic cables use light to send data over long distances very quickly. They also block all electrical interference from the big event speakers and massive LED screens, keeping the match from freezing or buffering.

(b) **Local Distribution: Switch to Gaming Consoles & Smart TVs**

Cable Choice: Cat6 or Cat6A Shielded Twisted Pair (STP)

Connector Choice: Standard Shielded RJ-45 Connector

Why this choice:

a. **Bandwidth & Speed:** Cat6 handles speeds up to **10 Gbps**, which is more than enough for high-end multiplayer gaming consoles and multiple high-definition 4K streaming displays.

b. **Low Latency:** Copper cables offer the absolute lowest latency over short distances (under 55 meters), which is critical for competitive gaming zones where milliseconds matter.

c. **Shielding (STP):** Event spaces are packed with interference from power generators, massive LED displays, and audio amplifiers. Shielded Cat6 cables and shielded RJ-45 metal connectors prevent this electrical noise from distorting your data packets and causing lag.

5. Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.



- **Cable Type:** This setup uses standard copper Cat6 Ethernet cables. You can easily spot the bright yellow cable and the matching grey cables plugged side-by-side.
- **Connector Type:** Each cable ends in a clear plastic RJ-45 connector, which clicks securely into the individual network ports.
- **Devices Connected:** This is the back panel of a **Wi-Fi internet router**. The yellow cable typically connects the router to the main internet source incoming from the wall or modem. The grey cables run directly out of the local LAN ports to share high-speed, wired internet with nearby desktop computers, smart TVs, or local network switches.